

Multimodal PDAC Metastatic Biomarker Discovery

The core objective is to develop a computational framework that integrates **Histopathology Foundation Model embeddings** with **Spatial Transcriptomics (ST)** to identify specific morphological features in primary pancreatic tumors that predict liver metastasis.

Crucially, the goal is to "augment" tissue embeddings during training so that, at inference time, the model can predict metastatic potential directly from H&E images without requiring ST data.

Dataset:

To train a model that "sees" transcriptomics in H&E, we need the following "Gold Standard" pairs:

- **Matched Tissue Triads:**
 - **Primary Tumor (PT):** H&E WSI + corresponding ST data.
 - **Liver Metastasis (HM):** H&E WSI + corresponding ST data (from the same patient if possible).
 - **H&E Only Samples:** A larger cohort of clinical H&E slides (PT and HM) without ST for large-scale validation and inference.
 - Does we have samples from the **same patient** for both the primary and the liver metastasis? (This is the "Gold Mine" for this specific research).
- **Single-Cell RNA-seq (scRNA-seq) Reference:** If your ST is "spot-level" (like 10x Visium), you need a scRNA-seq atlas of PDAC to **deconvolve** those spots into specific cell types (e.g., distinguishing malignant epithelial cells from fibroblasts).
- **Clinical Metadata:** Specifically, the **site of metastasis** (confirming liver) and, if possible, the "time to metastasis" for patients where you only have the primary tumor.
- **Gene Panel Coverage:** Ensure your ST data includes known PDAC metastatic markers (e.g., *S100A4*, *POSTN*, *MMP* family, or EMT-related genes). If these genes have "zero counts" across your slides due to low sequencing depth, the model won't have a "target" to learn.
- **Mitochondrial Read %:** High mitochondrial DNA in spots usually indicates dying cells. These spots should be filtered out, as their morphology is "distorted" and will confuse the Foundation Model.

Required Data:

Patient ID	PT Sample	HM Sample	Image Name	Clinical Data	WSI H&E Image
Patient 1	Available	NA	IU_PDA_T1	Available	Available
Patient 2	Available	Available	IU_PDA_HM2	Available	Missing
			IU_PDA_HM2_2	Missing	Missing
			IU_PDA_T2	Missing	Missing
Patient 3	NA	Available	IU_PDA_HM3	Missing	Missing
			IU_PDA_T3	Available	Available
Patient 4	Available	Available	IU_PDA_HM4	Available	Missing
			IU_PDA_T4	Missing	Available
Patient 5	NA	Available	IU_PDA_HM5	Available	Missing
Patient 6	Available	Available	IU_PDA_HM6,	Missing	Missing
			IU_PDA_T6	Available	Missing
Patient 7	NA	NA	NA	NA	NA
Patient 8	Available	Available	IU_PDA_HM8,	Missing	Missing
			IU_PDA_T8	Available	Missing
Patient 9	Available	Available	IU_PDA_HM9,	Missing	Missing
			IU_PDA_T9	Available	Missing
Patient 10	Available	Available	IU_PDA_HM10,	Missing	Missing
			IU_PDA_T10	Available	Missing
Patient 11	Available	Available	IU_PDA_HM11	Missing	Available
			IU_PDA_T11	Available	Available
Patient 12	Available	Available	IU_PDA_HM12,	Missing	Missing
			IU_PDA_T12	Available	Missing
Patient 13	NA	Available	IU_PDA_HM13	Available	Available

The below 7 .tiff files contain 4 image slides each, a mixture of known Zenodo samples and some entirely new samples. Also, 1 last .tif WSI is an unknown sample. The names are given below so we are not able to use these samples out of the 30 given samples.

1. 1_117.tiff
2. 2_119.tiff
3. 3_118.tiff
4. 120.tiff
5. V326.tiff
6. V327.tiff
7. V377.tiff
8. S21_23955_E7.tif

It is requested that all provided images be BIGTIFF-compatible, pyramidal, tiled TIFFs with JPEG compression, scanned at 40x and exported from the scanner's native software.

1. Clinical Data:

For reference, these are the data fields for clinical data:

Age	Gender	Ethnic	Location_in_pancreas
AJCC_stage	Tumor_Grade_Histology	R0_resection	Peripheral_Nerve_Invasion
Lymphovascular_Invasion	Nodes_positive	Neoadjuvant_Chemo	Treatment
Data_of_Surgery_start_point_in_survival_analysis	surgery_date	T_stage	N_stage
M_stage	nodes_pos	nodes_total	node_ratio

2. ST Data:

We have compiled a list of 62 metastasis marker genes through literature research; among those 6 genes were absent from the gene expression matrix in the given dataset:

- CDKN2A - Driver mutation
- TWIST1 - EMT / invasion
- MALAT1 - Epigenetic / ncRNA
- RACGAP1 - Prognostic signature
- RARRES3 - Prognostic signature
- IGF2BP3 - Signalling

The 6 missing genes are most probably caused by technical filtering during Seurat processing. Though their absence may affect the work, as genes are bundled up with other similar genes. However, for similar grouped genes, one missing gene would not significantly affect the model. If we can also get these, it would be great; otherwise, we can complete the experiment without these 6 genes.

Marker Gene Priority List for HM Samples (62 Genes):

Tier 1 — "Big Four" drivers (KRAS, TP53, SMAD4, CDKN2A): These are the most extensively validated. The accumulation of major driver alterations, including CDKN2A, KRAS, p53, and SMAD4, is inversely associated with disease-free and overall survival, and this altered gene combination is specifically linked to liver metastasis. SMAD4 loss in particular is the most strongly correlated single factor for metastatic recurrence.

Tier 2 — EMT/invasion genes: A global EMT expression signature identifies ~50 cancer cell clusters spanning the epithelial-mesenchymal continuum in both human and murine PDACs. Notably, total EMT suppression can *increase* liver metastasis via collective epithelial migration.

Tier 3 — Liver metastasis-specific: LITAF and SPRR1B show much higher expression in metastasis than primary PDAC samples, and high expression of these two genes is significantly associated with poor prognosis in the TCGA dataset. These came from a validated 11-gene malignant metastatic cell signature derived from scRNA-seq of synchronously resected PT+LM tissue.

Tier 4 — TME/stromal markers: Specific subtypes of stromal and immune cells critical to the formation of the pro-tumour microenvironment in metastatic lesions include RGS5+ cancer-associated fibroblasts, CCL18+ lipid-associated macrophages, S100A8+ neutrophils, and FOXP3+ regulatory T cells.

For the CITED4 gene specifically — this is a brand new one from Theranostics 2025 that might be worth knowing, as it was validated functionally via siRNA knockdown in the same snRNA-seq + spatial transcriptomics framework.

3. Mitochondrial Level:

The dataset shows 0% for percent_mito for each PT and HM sample.